

HST-Light: Hierarchical Spatio-Temporal Decoupling for Adaptive Traffic Signal Control

1st Quanwei Shen*

School of Computer Science and Technology
Chongqing University of Posts and Telecommunications
 Chongqing, China
 *s231231051@stu.cqupt.edu.cn

3rd Yuchi Liu

School of Computer Science and Technology
Chongqing University of Posts and Telecommunications
 Chongqing, China
 s231231046@stu.cqupt.edu.cn

5th Yu Kuang

School of Computer Science and Technology
Chongqing University of Posts and Telecommunications
 Chongqing, China
 s231231031@stu.cqupt.edu.cn

2nd Zhiqiang Zhao

School of Computer Science and Technology
Chongqing University of Posts and Telecommunications
 Chongqing, China
 zhaozq@cqupt.edu.cn

4th Fei Liu

School of Computer Science and Technology
Chongqing University of Posts and Telecommunications
 Chongqing, China
 s231231042@stu.cqupt.edu.cn

6th Zhu Xu

School of Computer Science and Technology
Chongqing University of Posts and Telecommunications
 Chongqing, China
 s231231076@stu.cqupt.edu.cn

Abstract—With urbanization accelerating and motor vehicle ownership growing, urban traffic congestion has become a bottleneck for urban development. Traditional traffic signal control systems, relying on fixed-time or actuated strategies, cannot adapt to dynamic traffic demands. In recent years, reinforcement-learning-based methods show potential but face three challenges: strong partial observability in traffic environments, tight spatiotemporal coupling, and dimensionality explosion in large-scale road networks. To solve these issues, this paper proposes HST-Light, an intelligent traffic signal control system based on multi-agent deep reinforcement learning. It constructs a hierarchical spatiotemporal attention mechanism, using spatiotemporal decoupling to handle traffic flow's temporal dynamics and spatial correlations. In the temporal dimension, it designs a Transformer encoder with adaptive windows; in the spatial dimension, it adopts a graph attention network. It also includes a dynamic memory bank module to enhance adaptability in unexpected scenarios. Experiments on the SUMO platform show that compared to conventional methods, HST-Light significantly improves key performance metrics, reducing average travel time by 15.45% and 19.13%. It demonstrates stronger robustness during peak hours and emergencies. This research offers an innovative solution for urban traffic management and insights for other spatiotemporal sequence decision-making problems.

Index Terms—Intelligent traffic signal control, Multi-scale attention, Multi-agent reinforcement learning, Spatiotemporal modeling, Real-time decision making, SUMO simulation

I. INTRODUCTION

A high-efficiency traffic signal control (TSC) system serves as the core to easing traffic gridlock. As stated in a World Bank report, the average yearly economic loss brought by urban traf-

fic congestion globally reaches 1-3% of GDP. In recent years, reinforcement learning (RL) has been extensively applied in the TSC field [1]. It is capable of interacting with, exploring and utilizing the environment, enabling agents to find superior strategies without manual pre-set hypotheses. A large number of studies [2-8] have shown that RL methods can reduce vehicle delays by 15-30% in simulation environments, and their performance is significantly better than traditional rule-based methods [9-10]. These studies include phase competition approaches [3], decentralized large-scale control [4], multi-agent frameworks [6], and fairness-aware methods [8]. Yet, current mainstream approaches remain scenario-dependent, which means that training and testing need to be performed in the same scenario (the scenario refers to a virtual city simulation environment with a set of intersections). When deployed to a new scenario, the environment needs to be rebuilt and retrained, which is costly. Therefore, rule-based systems such as SCATS or SCOOT are still widely used in cities, mainly because they can be easily adapted to new regions/cities. This leads to a key question: how to design a multi-intersection collaborative control system with strong generalization ability.

There are some solutions for single-agent settings: MetaLight [11] and GESA [12] achieve transfer capabilities through meta-learning and multi-scenario joint training, respectively, but they obviously ignore the cooperative relationship between multiple intersections. Designing a multi-agent RL model for multiple scenarios is challenging because the road network

structure and traffic dynamics of different scenarios are different, resulting in different environmental states and cooperation modes of each agent. Recent studies including Modlight [13] and MetaGAT [14] utilize Markov decision process (MDP) trajectories to aid agents in learning and distinguishing situational contexts, yet they still have drawbacks like unsteady training procedures and reduced performance when facing distinct scenarios.

To address these issues, we present **HST-Light**, a hierarchical spatiotemporal framework featuring three core innovative points: (1) **Spatiotemporal decoupling** that separates traffic control into temporal dynamics (modeled by adaptive time-window Transformers) and spatial correlations (captured via graph attention networks); (2) **Hierarchical decision-making**, where real-time intersection control operates at the bottom layer and regional coordination is handled at the upper layer through dynamic memory banks; (3) **Lightweight deployment** following a centralized-training-decentralized-execution paradigm, balancing scalability with distributed decision. This framework overcomes traditional methods' limitations in partially observable environments and reduces computational complexity in large-scale road networks.

II. RELATED WORK

Conventional traffic signal regulation approaches consist of fixed-timing plans (e.g., Webster's 1958 model [15]) and sensor-driven adaptive systems (e.g., SCOOT [9], SCATS [10]). While stable, these methods struggle with dynamic traffic flows and lack coordination capabilities. Reinforcement learning (RL) introduced adaptive control, with early works like PressLight [2] using Q-learning but facing challenges in computational cost and generalization.

Meta-reinforcement learning approaches like MetaLight [11] enabled cross-scenario adaptation, but ignored multi-intersection cooperation. Multi-agent methods such as MetaGAT [14] combined graph attention networks with meta-learning but suffered from training instability. GESA [12] proposed a general plug-in module for topology adaptation but had high computational complexity. Recent works like CLight[7] integrates contrastive learning into multi-agent reinforcement learning to boost cooperative representation for multi-intersection traffic signal control.

Transformer architectures have revolutionized decision systems, with Decision Transformer [17] reformulating RL as sequence modeling. In TSC, Transformer Light [18] applied offline learning, but faced data dependency issues. Other approaches like Efficient Pressure [16] focus on improving intersection efficiency through pressure-based optimization. Our work builds on these advances by introducing a **multi-scale spatiotemporal Transformer** framework that addresses spatiotemporal coupling and scalability challenges through hierarchical attention and dynamic memory mechanisms.

III. PRELIMINARY

We formulate TSC tasks as a Dec-POMDP. Each intersection i is an agent with state $s_i^t = [q_i^t, o_i^t, \phi_i^t, f_i^t]$ (queue length,

occupancy, phase, flow), observation o_i^t including neighbor states, action $a_i^t \in \{0, 1\}^k$ (phase selection), and reward $r_i^t = -\alpha q_i^t - \beta w_i^t + \gamma d_i^t$. The traffic road system is denoted as a graph $G = (V, E)$ with edges $e_{ij} = (l_{ij}, c_{ij}, \tau_{ij})$. Phase transitions follow constraint matrix P_i . A typical intersection is shown in Fig. 1, and the corresponding phase is shown in Fig. 2.

The objective is to find optimal policy π^* maximizing:

$$\mathbb{E}_\pi \left[\sum_{t=0}^T \gamma^t \left(\sum_{i=1}^N r_i^t + \lambda R_g^t \right) \right], \quad (1)$$

where R_g^t is a global metric, λ coordinates local/global optimization, and γ discounts future rewards.

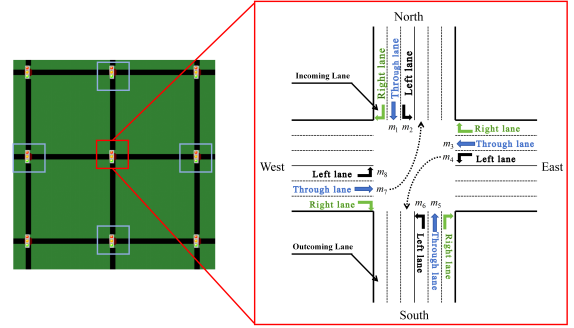


Fig. 1. A basic simulation scene and a typical intersection configuration. The red frame denotes the core intersection under control, and the blue frames stand for its four closest adjacent intersections.

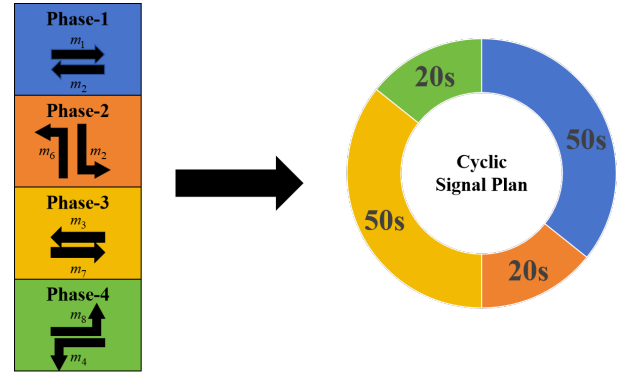


Fig. 2. Eight typical signal phases.

IV. METHODS

HST-Light employs a hierarchical spatiotemporal Transformer architecture for multi-intersection collaborative control. The framework comprises: (1) Perception layer (GAT networks fusing local observations and neighborhood information), (2) Spatiotemporal optimization layer (dynamic time windows for historical patterns + graph convolution for topological constraints), (3) Multi-scale fusion layer (dynamic weighted combination of local/regional/global features), and

(4) Decision layer (generating signal actions and phase difference coordination). The system architecture is shown in Fig. 3.

A. Time Dimension Optimization

Temporal optimization uses adaptive windows: $\tau^* = \arg \min_{\tau} \|\hat{W}_t - W_t\|^2 + \lambda \|\nabla \tau\|^2$. Composite positional encoding $\text{PE} = \text{LayerNorm}(\text{PE}_t + \text{PE}_p + \text{PE}_e)$ captures instantaneous, periodic, and event-based patterns. Sparse attention $A_{ij} = Q_i K_j^T / \sqrt{d}$ if $M_{ij} = 1$, else $-\infty$, reduces computation.

B. Spatial Co-Optimization

Spatial modeling employs GCN: $H^{(l+1)} = \sigma(\hat{D}^{-1/2} \hat{A} \hat{D}^{-1/2} H^{(l)} W^{(l)})$. Influence factor $\phi_{ij} = \kappa \exp(-\|p_i - p_j\| - q_j^{\text{out}}/c_{ij})$ incorporates distance and capacity. Attention weights $\alpha_{ij} = \text{softmax}(Q_i K_j^T / \sqrt{d} + \log \phi_{ij})$ explicitly encode topology.

C. Multi-scale Feature Dynamic Fusion

The fusion architecture employs a hierarchical process where local features f_L capturing real-time vehicle dynamics are first extracted via three dilated convolutions ($d = 1, 2, 4$), while temporal features $h_{\text{time}}^{\text{time}}$ and spatial features $h_{\text{space}}^{\text{space}}$ are fused through cross-attention to generate regional features f_R , followed by graph attention pooling aggregating network-wide regional features into global features f_G , and finally a dynamic gating mechanism:

$$g_t = \sigma(W_g[f_L, f_R, f_G]), \quad (2)$$

computes normalized fusion weights $g_t^L + g_t^R + g_t^G$ to produce the multi-granularity representation f^{fusion} through weighted summation of MLP-projected features at all scales:

$$f^{\text{fusion}} = \sum_{k \in \{L, R, G\}} g_t^k \cdot \text{MLP}(f_k). \quad (3)$$

D. Actor-Critic

Based on the Actor-Critic framework, the loss function $L = \lambda_1 L_{\text{policy}} + \lambda_2 L_{\text{value}} + \lambda H(\pi)$, comprising policy loss, value loss, and a policy entropy regularization term. Hyperparameters λ_1 and λ_2 control the balance between these components. The policy network employs a Gaussian distribution $\pi(a|s) = \mathcal{N}(\mu(s), \sigma^2(s))$ for continuous action spaces. For discrete phase selection, Gumbel-Softmax approximation is used for sampling.

E. Multi-stage Joint Training

Multi-scenario co-training enhances the generalization of traffic signal control models by simultaneously optimizing policies across diverse scenarios (e.g., peak hours, off-peak periods, and accident conditions). During training, the algorithm aggregates multi-scale traffic features from different scenarios and employs scenario-adaptive fusion mechanisms. By dynamically balancing learning weights across scenarios and incorporating a shared memory module for cross-scenario knowledge transfer, the model learns robust representations while maintaining scenario-specific adaptability.

This approach ensures consistent performance under varying traffic conditions without requiring separate models for each scenario. The training procedure is illustrated in Algorithm 1.

Algorithm 1: HST-Light Training Process

- 1) **Input:** Road network graph $G = (V, E)$, hyperparameters (α : Learning rate, γ : Discount factor, λ_1, λ_2 : Weights)
 - 2) **Output:** Optimized policy networks π_{θ} for all intersections
 - 3) **Initialize:** Actor network π_{θ} , Critic network V_{ϕ} , Replay buffer $\mathcal{D} = \emptyset$, Dynamic memory bank $\mathcal{M} = \emptyset$, Scenario set $\mathcal{S} = \{\text{Grid4x4}, \text{Arterial4x4}, \dots\}$
 - 4) **for** episode = 1 to E **do**
 - a) Randomly select scenario $s \in \mathcal{S}$
 - b) Initialize traffic environment with G and s
 - c) Get initial observations o_i^0 for all $i \in V$
 - d) **for** $t = 0$ to $T - 1$ **do**
 - e) **Hierarchical Decision-Making**
 - f) **for** each intersection $i \in V$ (parallel) **do**
 - i) $z_i^t \leftarrow \text{MLP}(o_i^t) \oplus \text{GAT}(\{o_j^t \mid j \in \mathcal{N}(i)\})$
 - ii) $m_i^t \leftarrow \text{Attention}(\mathcal{M}, z_i^t)$
 - iii) $a_i^t \sim \pi_{\theta}(z_i^t \oplus m_i^t)$
 - iv) $\text{Execute_Phase_Transition}(a_i^t)$ s.t.
 - v) Observe rewards $r_i^t = -(\alpha q_i^t + \beta w_i^t + \gamma d_i^t)$ and new states o_i^{t+1}
 - vi) Store transition (o^t, a^t, r^t, o^{t+1}) in $\mathcal{D}$
 - g) **end for**
 - h) **if** $t \bmod \text{update_interval} = 0$ **then**
 - i) Sample batch $\mathcal{B} \sim \mathcal{D}$
 - ii) **Multi-Scale Feature Fusion**
 - iii) **for** each $(o_i, a_i, r_i, o_i') \in \mathcal{B}$ **do**
 - A) $f_L \leftarrow \text{Dilated_Conv1D}(o_i, [1, 2, 4])$
 - B) $h_{\text{time}} \leftarrow \text{AdaptiveWindowTransformer}(o_i)$
 - C) $h_{\text{space}} \leftarrow \text{GAT} + \text{SpatialAttention}(o_i)$
 - D) $f_R \leftarrow \text{CrossAttention}(h_{\text{time}}, h_{\text{space}})$
 - E) $f_G \leftarrow \text{GraphAttentionPooling}(\{f_R \mid \forall i \in G\})$
 - F) $g_t \leftarrow \sigma(\mathbf{W}_g[f_L, f_R, f_G])$
 - G) $f^{\text{fusion}} \leftarrow \sum_{k \in \{L, R, G\}} g_t^k \cdot \text{MLP}(f_k)$
 - iv) **end for**
 - v) **Loss Computation**
 - vi) $\mathcal{L}_{\text{actor}} \leftarrow -E[\log \pi_{\theta}(a_i | z_i') \cdot A(z_i')]$
 - vii) $\mathcal{L}_{\text{critic}} \leftarrow E[(r_i + \gamma V_{\phi}(z_i') - V_{\phi}(z_i))^2]$
 - viii) $\mathcal{L}_{\text{entropy}} \leftarrow E[\pi_{\theta} \log \pi_{\theta}]$
 - ix) $\mathcal{L}_{\text{total}} \leftarrow \mathcal{L}_{\text{actor}} + \lambda_1 \mathcal{L}_{\text{critic}} + \lambda_2 \mathcal{L}_{\text{entropy}}$
 - x) Update $\theta \leftarrow \theta - \alpha \nabla \mathcal{L}_{\text{total}}$
 - xi) Update $\phi \leftarrow \phi - \alpha \nabla \mathcal{L}_{\text{critic}}$
 - xii) $\mathcal{M} \leftarrow \text{LRU_Update}(\mathcal{M}, \{z_i \mid \forall i \in V\})$
 - i) **end if**
 - j) **end for**
 - 5) **end for**
 - 6) **return** θ^*
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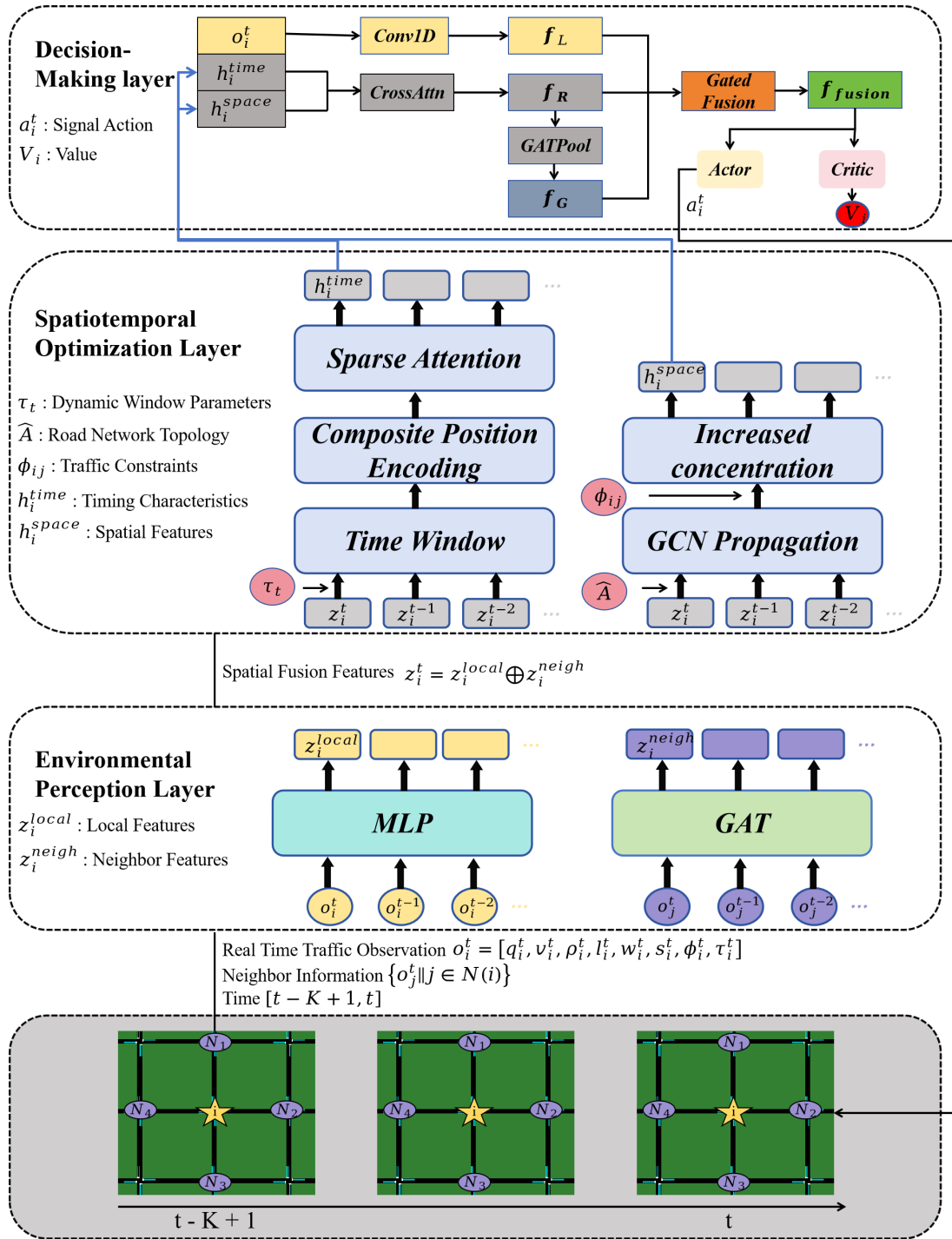


Fig. 3. The HST-Light system architecture. The framework comprises: (1) Perception layer (GAT networks fusing local observations and neighborhood information), (2) Spatiotemporal optimization layer (dynamic time windows for historical patterns + graph convolution for topological constraints), (3) Multi-scale fusion layer (dynamic weighted combination of local/regional/global features), and (4) Decision layer (generating signal actions and phase difference coordination).

V. EXPERIMENTS

A. Datasets

We carried out simulations with the Simulation of Urban Mobility (SUMO) as the traffic simulation tool, where each simulation episode lasts 3600 seconds. To ensure stable traffic

flow before data collection, we implement a warm-up period of 300 seconds at the beginning of each simulation episode. During this warm-up phase, vehicles are introduced into the network without any performance metrics being recorded, allowing the traffic system to reach a steady state. This approach

eliminates transient effects and ensures that the collected data reflects stable operational conditions.

To simulate realistic traffic conditions, we define distinct vehicle arrival rates and distribution types for peak and off-peak periods. During peak hours (7:00-9:00 and 17:00-19:00), vehicle arrival follows a Poisson distribution with mean arrival rates ranging from 600 to 1200 vehicles per hour (vph) depending on the scenario complexity. For off-peak periods, we employ a uniform distribution with arrival rates between 200 and 400 vph. These parameters are calibrated based on real-world traffic data from urban traffic management systems to ensure simulation fidelity.

Our multi-scenario co-training framework incorporates seven distinct traffic scenarios [12][20], with four primary scenarios reported in the results and three additional scenarios (Fenglin, Nanshan and Grid 5×5) used to enrich meta-training diversity. To enable efficient co-training across scenarios, we employ the Generalized Policy Interface (GPI) module from GESA, which automatically standardizes observation and action spaces across all intersections without requiring manual labeling. The GPI accomplishes this by converting intersections with diverse structures into a standardized 4-leg expression through relative angles and masking technologies. See [12] for implementation specifics. TABLE I summarizes the key characteristics of each scenario.

TABLE I
STATISTICS OF THE SCENARIOS

Scenarios	Country	Type	#Total Int.
Grid 4×4	synthetic	region	16
Avenue 4×4	synthetic	region	16
Grid 5×5	synthetic	region	25
Cologne8	Germany	region	8
Ingolstadt21	Germany	region	12
Fenglin	China	corridor	7
Nanshan	China	region	28

B. Baselines

To fully confirm the validity of the proposed approach, we selected four categories of methods for thorough comparison: Traditional methods, Multi-agent-based methods, Meta Learning-based methods, and the hybrid type integrating Meta Learning and Multi-agent.

Conventional methods:

- **Fixed Time Control (FTC)** with random offset cycles through each phase in a loop, adopting a preset phase duration.
- **MaxPressure** [19] is a decentralized controller using local queue feedback for signalized arterial networks with two real-time applicable variations.

Multi-agent methods:

- **MPLight** is a decentralized RL method with parameter sharing integrating pressure concept for large-scale traffic signal control using feasible data.

- **IPPO** adopts IDQN-based deep neural network with Atari default hyperparameters for traffic signal control tasks via Preferred RL.
- **rMAPPO** adopts centralized value function inputs with agent-specific and global features, shares parameters among homogeneous agents, and applies PPO's core mechanisms for cooperative multi-agent tasks.
- **CoLight** [5] uses graph attention networks for index-free multi-agent traffic signal cooperation.

MetaLearning and Single-agent methods:

- **MetaLight** [11] integrates FRAP++ with individual-global adaptations for value-based meta-RL traffic signal control.
- **GESA** [12] unifies intersections via plug-in module, adopts unified state-action spaces, and modifies FRAP for multi-scenario co-training in traffic control.

MetaLearning and Multi-agent methods:

- **MetaGAT** [14] combines context-based meta-RL with GAT, decouples task inference and control, and uses multi-task training for multiple traffic signal control.
- **HSTLight(ours)**, which adopts a three-level progressive architecture of "spacetime decoupling multi-scale dynamic fusion-curricular collaborative training", decouples spacetime feature processing, dynamically fuses multi-scale road network information through gating, and hierarchically optimizes the collaborative decision-making of intelligent agents based on curriculum learning strategies.

C. Evaluation Metrics

In line with the majority of contemporary research, we include **mean travel duration** in our assessment criteria, which stands for the average time each vehicle consumes from accessing to departing the emulation scene. That said, using mean travel duration alone cannot authentically reflect the actual traffic condition—for instance, during extreme gridlock, incoming vehicles are blocked from entering the scene, leading to a relatively low mean travel duration even amid heavy congestion. Hence, we further add **mean latency duration** as an assessment metric, defined as the delay induced by signal-regulated intersections and traffic jams.

D. Results

Performance Enhancement: HST-Light achieves significant improvements across scenarios (Tables II-III). In Grid 4×4 and Arterial 4×4 , trip time reduces by **29.4%** and **28.3%**, delay time by **42.4%** and **43.2%**. Key innovations: dynamic window LSTM for temporal patterns, topology encoder for spatial relationships, and three-level feature fusion integrating microscopic queue data with macroscopic congestion patterns.

Generalization: Maintains **5.6%** trip time advantage in Ingolstadt21 but shows **1.5%** decline in Cologne8 due to limitations in ultra-large network topology modeling. The performance variation highlights the model's sensitivity to network scale, with ultra-large networks exceeding current capacity for long-range spatial dependencies. Delay time improvement

TABLE II

MEAN TRAVEL DURATION IN CROSS-SCENARIO DEPLOYMENT. TOP PERFORMERS MARKED IN **BOLD** AND RUNNERS-UP IN UNDERLINES (PERCENT GAIN RELATIVE TO THE SECOND-PLACE METHOD). OUR PROPOSED STRATEGY ALSO HITS THE TOP SPOT, WITH AN OVERALL IMPROVEMENT RATE OF **15.45%**.

Methods	Avg. Trip Time (seconds)			
	Grid4×4 (seen)	Arterial4×4 (seen)	Ingolstadt21 (seen)	Cologne8 (seen)
FTC	206.68	828.38	319.41	124.4
MaxPressure	175.97	686.12	375.25	95.96
MPLight	179.51	541.29	319.28	98.44
IPPO	167.62	431.31	379.22	90.87
rMAPPO	164.96	565.67	453.61	97.68
CoLight	163.52	409.93	337.46	89.72
MetaLight	169.21	381.77	337.46	91.57
GESA	<u>161.33</u>	393.57	292.26	<u>90.59</u>
MetaGAT	165.23	<u>374.80</u>	<u>290.73</u>	90.74
Ours	113.87(+29.4%)	268.91(+28.3%)	274.51(+5.6%)	91.07(-1.5%)

TABLE III

MEAN LATENCY DURATION IN CROSS-SCENARIO DEPLOYMENT. TOP PERFORMERS MARKED IN **BOLD** AND RUNNERS-UP IN UNDERLINES (PERCENT GAIN RELATIVE TO THE SECOND-PLACE METHOD). OUR PROPOSED STRATEGY ALSO HITS THE TOP SPOT, WITH AN OVERALL IMPROVEMENT RATE OF **19.13%**.

Methods	Avg. Delay Time (seconds)			
	Grid4×4 (seen)	Arterial4×4 (seen)	Ingolstadt21 (seen)	Cologne8 (seen)
FTC	94.64	1234.30	183.70	62.38
MaxPressure	64.01	952.53	275.36	31.93
MPLight	67.52	1083.18	185.04	34.38
IPPO	56.38	914.58	247.68	26.82
rMAPPO	53.65	1185.23	72.20	33.37
CoLight	51.58	776.61	226.06	25.56
MetaLight	57.56	862.32	164.80	27.61
GESA	<u>49.60</u>	775.22	209.57	<u>26.50</u>
MetaGAT	53.20	<u>772.36</u>	<u>176.86</u>	26.85
Ours	28.56(+14.3%)	437.70(+20.1%)	146.02(+17.7%)	30.82(-19.5%)

(19.13%) exceeds trip time (15.45%), validating precise phase timing control that specifically mitigates congestion-induced delays. Outperforms MetaGAT in training efficiency via parallel data collection, achieving $10.3\times$ acceleration through distributed simulation. The model demonstrates robust performance across diverse scenario types, with consistent improvements in region-based networks and competitive performance in corridor-based scenarios, underscoring the effectiveness of the hierarchical spatiotemporal attention mechanism.

Ablation Studies: Systematic ablation across four scenarios validates component contributions (Fig. 4). **Temporal module** removal causes **50.5%** degradation in Grid4×4 and **47.8%** in Arterial4×4. **Spatial module** removal shows severe impact in Grid4×4 (**121.7%**) but minimal in Arterial4×4 (**2.3%**), highlighting topology sensitivity. **Multiscale fusion** removal causes **96.6%** drop in Arterial4×4, critical for complex interactions. Joint removal reveals synergistic effects (221.7% degradation in Grid4×4), validating spatiotemporal joint optimization. Parameter $\alpha = 1.0$ remains stable, demonstrating

robustness.

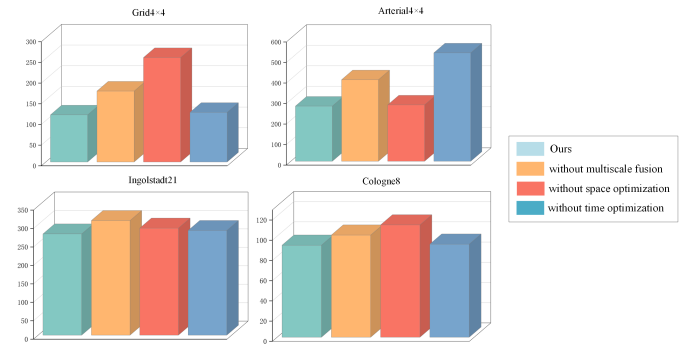


Fig. 4. Ablation experiments were conducted on every core component of HST-Light. The optimal scheme we derived secured the minimum travel time across all ablation tests, thus verifying the validity of our designed module configurations.

VI. CONCLUSION

HST-Light achieves **29.4%** trip time reduction and **43.2%** delay time improvement in Grid4×4/Arterial4×4 scenarios. Key innovations: dynamic window LSTM for temporal patterns, topology-aware spatial encoder, and three-level feature fusion. Ablation studies reveal synergistic effects (49.5% additional gain) validating spatiotemporal joint optimization. Limitations include **1.5%** performance gap in ultra-large networks (Cologne8). Engineering insights: (1) topology-specific module configurations, (2) stable parameter $\alpha = 1.0$, (3) 10.3× training acceleration via parallel simulation achieved on hardware with NVIDIA RTX 4090 GPU (24GB VRAM) and Intel Core i9-13900K CPU (5.8 GHz clock frequency). Future work: hierarchical attention for complex networks.

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